Joskev Developers SmartPOS

Complete User & Administrator Manual

Powered by Joskev Developers

Document Version: 1.0
Platform: Android SmartPOS
Recommended Device: Android 12+, 4 GB RAM, 64 GB storage, good-quality camera

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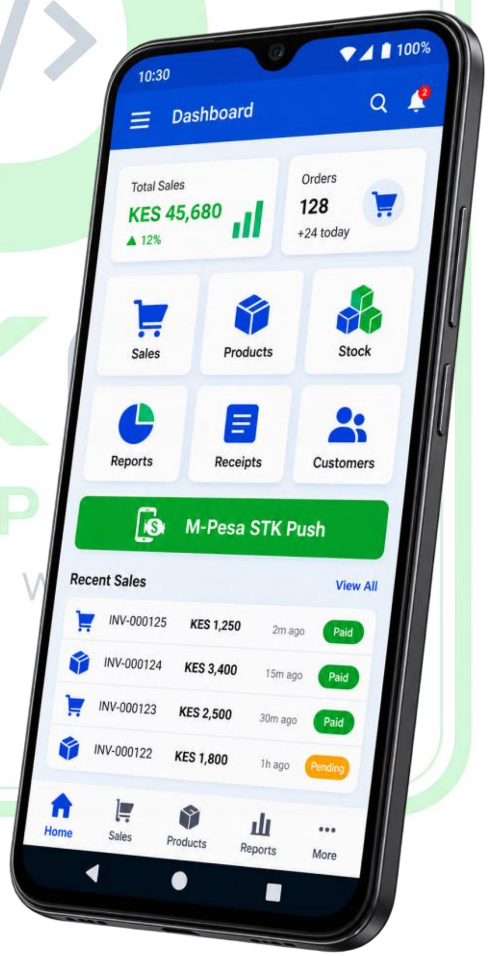

1. Introduction

The **Joskev Developers SmartPOS** is an Android-based Point of Sale system designed to help businesses manage daily retail operations from a single SmartPOS device.

The system combines:

- Sales management
- Product management
- Stock management
- Barcode scanning
- M-Pesa STK Push
- Receipt printing
- PDF sales receipts and references
- Finance and reporting
- Cashier management
- Administrator management
- Offline-first SQLite storage
- Automatic cloud backup
- Backup history
- Secure local data management

The system is designed for businesses that need a POS that can continue operating even when internet connectivity is temporarily unavailable.

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2. SmartPOS Overview

The SmartPOS is designed around an **offline-first architecture**.

This means the local database is not dependent on the internet for every operation.

The primary working database is the local **SQLite database**.

The general operating model is:

SmartPOS → Local SQLite Database → Business Operations

and, when internet connectivity is available:

Local Database → Backup System → Cloud Backup

This allows the business to continue normal local operations during temporary internet outages while still benefiting from cloud backup when connectivity returns.

Example

Suppose the SmartPOS loses internet connectivity on Monday.

The business can continue performing supported local operations.

The device can remain offline for several days.

When internet connectivity returns, the backup system can identify pending backup information and synchronize it.

The administrator can then check the backup history and see the latest successful backup date.

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3. Main Features

3.1 M-Pesa STK Push

The system can initiate an M-Pesa payment request to a customer's phone.

The customer receives an STK prompt and confirms the payment using their M-Pesa PIN.

3.2 Sales Management

The POS allows cashiers to:

- Select products
- Scan barcodes
- Set quantities
- Review totals
- Process payments
- Complete sales
- Generate receipts
- Maintain sales records

3.3 PDF Sales Receipts and References

Completed sales can be represented using PDF documents or sales references.

These can be useful for:

- Customer records
- Business records
- Sales verification
- Reconciliation
- Management review

3.4 Product and Stock Management

Administrators can maintain:

- Product names
- Barcodes
- Prices
- Quantities
- Stock information

Stock information can be used to monitor available inventory and identify products that need replenishment.

3.5 Barcode Scanning

The SmartPOS uses the device camera to scan supported barcodes.

This allows a cashier to identify products quickly instead of manually searching for them.

3.6 Receipt Printing

The POS can work with a compatible receipt printer.

A typical workflow is:

Sale completed → Receipt generated → Printer selected → Receipt printed

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3.7 Offline-First SQLite Database

The local SQLite database provides the primary operational storage.

This means the system can maintain important local information even when internet connectivity is unavailable.

3.8 Automatic Cloud Backup

When internet connectivity is available, pending backup information can be uploaded automatically.

This reduces the need for the cashier to manually perform every backup.

3.9 Backup History

The backup system should maintain information about successful backups.

The administrator can use backup history to determine:

- Whether a backup succeeded
- The last successful backup date
- Whether there are pending backups

3.10 Finance and Reports

The finance/reporting functionality provides management with information about business activity.

Depending on the installed version, this may include:

- Sales
- Revenue
- Transaction history
- Financial summaries
- Sales references

3.11 Cashier and Administrator Management

The system separates normal cashier operations from administrative functions.

This improves accountability and security.

3.12 Secure Local Database

Operational information is maintained in the local database.

Administrators should protect the physical device and account credentials to prevent unauthorized access.

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4. Recommended Hardware

For the best experience, the SmartPOS should use the following specifications.

Specification	Recommendation
Operating System	Android 12 or newer
RAM	4 GB or more
Storage	64 GB or more
Camera	Good-quality camera
Barcode Scanning	Autofocus camera recommended
Printer	Compatible thermal receipt printer
Internet	Wi-Fi or mobile data
Power	Reliable power/battery

4.1 Android 12+

Android 12 or newer is recommended for compatibility, security and a modern SmartPOS environment.

4.2 4 GB RAM

At least 4 GB RAM is recommended for smooth operation of the POS, camera scanner and other Android services.

4.3 64 GB Storage

64 GB storage provides sufficient room for:

- Application files
- Local database
- PDF references
- System data
- Normal Android operation

4.4 Good Camera

A good camera is particularly important for barcode scanning.

The camera should ideally have:

- Autofocus
- Good image clarity
- Good performance in indoor lighting

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5. System Architecture

The SmartPOS uses several major components.

5.1 Android Application

The Android application provides the user interface.

Cashiers and administrators interact with the POS through the Android interface.

5.2 Local SQLite Database

SQLite stores local operational information.

This is the primary database for offline-first operation.

Examples of information stored locally may include:

- Products
- Stock
- Sales
- Users
- Transaction information
- Backup-related information

5.3 Barcode Scanner

The barcode scanner uses the Android camera to recognize supported barcode formats.

5.4 M-Pesa Integration

M-Pesa communication is handled through the configured payment integration.

The POS can initiate an STK Push and receive the resulting transaction status.

5.5 Receipt Printer

The SmartPOS can communicate with a supported receipt printer.

Network printing may require the printer and SmartPOS to be connected to the appropriate local network.

5.6 Cloud Backup

The cloud backup system provides an additional copy of important business information.

The cloud should be treated as a backup destination rather than replacing the local database as the primary operating database.

6. First-Time Setup

Before using the SmartPOS for live business, perform the following setup.

Step 1 — Prepare the Device

Confirm that the SmartPOS has:

- Android 12+
- At least 4 GB RAM
- At least 64 GB storage
- A working camera
- Adequate battery/power

Step 2 — Install the Application

Install the SmartPOS application on the Android device.

Open the application after installation.

Step 3 — Grant Permissions

Android may request permissions such as camera access.

Allow camera access because barcode scanning requires the camera.

Step 4 — Initialize the Database

During the first launch, the application initializes the local SQLite database.

Allow the initialization process to complete.

Do not force-close the application during database initialization.

Step 5 — Create Administrator Account

Set up the administrator account.

The administrator account should be protected with a strong password.

Step 6 — Create Cashier Accounts

Create separate accounts for staff members who will operate the POS.

Avoid sharing one account between multiple employees.

Step 7 — Add Products

Enter the initial product catalogue.

Include:

- Product name
- Barcode
- Selling price
- Opening stock

Step 8 — Configure Printer

Connect the receipt printer.

Perform a test print.

Step 9 — Configure M-Pesa

Verify the M-Pesa configuration before accepting live payments.

Perform a controlled test transaction where appropriate.

Step 10 — Test Backup

Perform a backup test.

Confirm that the backup succeeds.

Check that the backup history shows the successful backup date.

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7. User Accounts and Roles

7.1 Administrator

The administrator is responsible for system management.

Administrator responsibilities include:

- Product management
- Stock management
- Cashier management
- Finance reports
- Sales review
- Backup monitoring
- Security
- System configuration

7.2 Cashier

The cashier is responsible for daily sales.

Typical cashier activities include:

- Logging in
- Scanning products
- Selecting products
- Adjusting quantities
- Processing sales
- Initiating M-Pesa payments
- Printing receipts
- Generating sales references

7.3 Account Security

Each user should have their own account.

Do not share administrator credentials.

Do not write administrator passwords where customers or unauthorized staff can see them.

8. Product Management

Product management is used to create and maintain products.

Adding a Product

Typical procedure:

1. Open Product Management.
2. Select **Add Product**.
3. Enter the product name.
4. Enter or scan the barcode.
5. Enter the selling price.
6. Enter the opening quantity.
7. Save the product.
8. Confirm that the product appears in the product list.

Editing a Product

When a product changes:

1. Find the product.
2. Open the product information.
3. Update the required field.
4. Save the changes.
5. Confirm the updated information.

Product Barcode

The barcode should be unique to the product where possible.

Incorrect barcode information can cause scanning to select the wrong product.

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9. Stock Management

Stock management helps the business know how much inventory is available.

Receiving New Stock

When goods arrive:

1. Confirm the physical quantity.
2. Locate the product.
3. Increase the stock quantity.
4. Save the adjustment.
5. Confirm the new quantity.

Monitoring Stock

Administrators should regularly check:

- Low-stock products
- Fast-moving products
- Unexpected stock changes
- Physical stock versus system stock

Stock Accuracy

The system is only as accurate as the information entered into it.

If a cashier sells five items but the sale is not recorded, the stock information may become inaccurate.

Therefore:

Every sale should be recorded through the POS.

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10. Barcode Scanning

Barcode scanning is designed to make sales faster.

Normal Scanning Procedure

1. Open the barcode scanner.
2. Grant camera permission if requested.
3. Place the barcode inside the scanning area.
4. Hold the device steady.
5. Keep the barcode reasonably well lit.
6. Allow the scanner to recognize the barcode.
7. Confirm the matched product.
8. Continue scanning other products.

Improving Scanning Speed

For better scanning:

- Keep the camera lens clean.
- Avoid extreme darkness.
- Avoid strong reflections.
- Hold the barcode relatively straight.
- Do not move the device too quickly.
- Use a good-quality camera.

Barcode Not Found

If the scanner recognizes a barcode but no product is found:

1. Check the barcode.
2. Open Product Management.
3. Confirm that the barcode exists.
4. Confirm that the barcode belongs to the correct product.
5. Save/update the product if necessary.

11. Sales Management

The sales process should be simple and consistent.

Standard Sale

Step 1

Cashier logs in.

Step 2

Scan or select the first product.

Step 3

Adjust quantity if necessary.

Step 4

Add additional products.

Step 5

Review:

- Products
- Quantities
- Prices
- Total

Step 6

Select payment method.

Step 7

Complete the payment.

Step 8

Save/complete the sale.

Step 9

Print the receipt or generate the PDF reference.

Before Payment

Always check:

- Correct product
- Correct quantity
- Correct price
- Correct total

After Payment

Confirm that the payment was successfully processed before treating the sale as completed.

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12. M-Pesa STK Push

M-Pesa STK Push is used to request payment from a customer's phone.

Standard Process

1. Confirm the customer's purchase.
2. Confirm the total amount.
3. Enter or confirm the customer's M-Pesa phone number.
4. Initiate STK Push.
5. Customer receives the M-Pesa prompt.
6. Customer checks the amount.
7. Customer enters their M-Pesa PIN.
8. Wait for the transaction result.
9. Confirm successful payment.
10. Complete the sale.
11. Keep the transaction/reference information.

Important Payment Rule

Never rely only on the customer's statement:

"I have paid."

The cashier should wait for the appropriate payment confirmation/result from the payment system.

Failed STK Push

If the STK Push fails:

1. Check the phone number.
2. Check network connectivity.
3. Confirm M-Pesa service availability.
4. Check the payment configuration.
5. Review the returned transaction result.
6. Retry only when appropriate.

Avoid creating duplicate payments accidentally.

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13. Receipt Printing

The SmartPOS can print physical receipts using a compatible thermal printer.

Printer Setup

1. Power on the printer.
2. Load thermal paper.
3. Connect the printer using the configured method.
4. Make sure the printer is available to the SmartPOS.
5. Discover/select the printer where supported.
6. Perform a test print.

Printing a Sale

After a completed sale:

1. Open the completed transaction.
2. Select the print option.
3. Confirm the printer.
4. Print.
5. Check that the receipt is readable.

Printer Troubleshooting

Check:

- Printer power
- Paper
- Paper orientation
- Network connection
- Printer selection
- Printer availability

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14. PDF Sales Receipts and References

PDF sales documents provide a digital transaction record.

They may contain information such as:

- Sale reference
- Date
- Products
- Quantities
- Prices
- Total
- Payment information

Generating a PDF

1. Complete the sale.
2. Open the completed transaction.
3. Select the PDF/reference option.
4. Generate the document.
5. Review the information.

6. Store or share it where required.

Why PDF References Are Useful

They can help with:

- Customer queries
- Business records
- Reconciliation
- Sales verification
- Management review

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15. Finance and Reports

Finance and reports provide management information.

Possible information includes:

- Number of sales
- Sales totals
- Revenue
- Transaction history
- Financial summaries

Daily Review

Administrators should review:

- Total sales
- Revenue
- Payment records
- Unusual transactions

Reconciliation

POS records should be compared with actual payment records.

For example:

POS sales → Cash received → M-Pesa received → Other payment methods

Differences should be investigated.

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16. Offline-First SQLite Operation

One of the most important characteristics of the SmartPOS is its offline-first design.

What Offline-First Means

The POS does not depend on a constant internet connection for all local operations.

The local SQLite database remains available on the device.

During an Internet Outage

The system can continue supported local operations.

For example:

- Product lookup
- Barcode scanning
- Local sales
- Local stock management
- Local transaction storage

Online services may not work until connectivity returns.

What May Require Internet

Depending on the implementation:

- M-Pesa communication
- Cloud backup
- Online synchronization

may require internet connectivity.

Example: One Week Offline

Imagine the shop loses internet on Monday.

The SmartPOS continues storing local business activity.

The shop operates Tuesday, Wednesday, Thursday and the following days.

When internet returns:

1. The application detects connectivity.
2. Pending backup information is identified.
3. Pending information can be uploaded.
4. Backup history is preserved.
5. The administrator can see the latest successful backup date.

The cashier should not need to recreate the week's local transactions simply because the internet was unavailable.

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17. Automatic Cloud Backup

The backup system provides additional protection for business data.

Normal Backup Flow

Local SQLite

↓

Pending backup data

↓

Internet becomes available

↓

Cloud backup

↓

Backup history updated

Automatic Backup

When connectivity is available, the application can automatically synchronize pending backup information.

The administrator should periodically verify that this process is succeeding.

Manual Backup

Where the manual backup function is available:

1. Open **Backup & Reports**.
2. Check database status.
3. Start the backup.
4. Wait for completion.
5. Check the result.
6. Confirm the backup date.

Backup Rule

Never assume a backup succeeded.

Always verify the backup status.

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18. Backup History and Recovery

Backup history provides evidence of previous successful backups.

The administrator should be able to identify:

- Last successful backup
- Backup date

- Pending backup status
- Historical backup activity

After Several Days Offline

If the POS has been offline for several days:

1. Reconnect the device to the internet.
2. Open the SmartPOS.
3. Allow synchronization/backup to operate.
4. Do not immediately clear application data.
5. Check backup status.
6. Confirm the latest successful backup date.
7. Preserve the backup history.

Before Device Replacement

Before replacing a SmartPOS device:

1. Perform a backup.
2. Confirm success.
3. Record the last successful backup date.
4. Keep the original device/database safe.
5. Configure the replacement device.
6. Follow the approved recovery/migration process.
7. Verify business data before retiring the old device.

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19. Cashier Daily Procedure

Opening

The cashier should:

- Power on the SmartPOS.
- Check battery/power.
- Open the application.
- Log in.
- Confirm the POS is ready.

- Check the scanner.
- Check the receipt printer.
- Report any problem before starting sales.

During Business

The cashier should:

- Record every sale.
- Scan products correctly.
- Check quantities.
- Confirm prices.
- Confirm totals.
- Verify M-Pesa results.
- Issue receipts.
- Avoid sharing login credentials.

Closing

At the end of the shift:

- Complete outstanding transactions.
- Review sales.
- Confirm payment information.
- Report discrepancies.
- Ensure required receipts/references are available.
- Log out where required.
- Secure the device.

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20. Administrator Daily Procedure

The administrator should regularly:

- Review sales.
- Review revenue.
- Check stock.
- Check unusual transactions.
- Check cashier activity.

- Check printer status.
- Check M-Pesa-related transactions.
- Check backup status.
- Check the last successful backup date.

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21. Security and Data Protection

Security is an important part of POS operation.

Account Security

Use:

- Individual user accounts
- Strong passwords
- Restricted administrator access

Physical Security

The SmartPOS should not be left unattended in an unsecured location.

Database Security

Do not manually modify SQLite database files unless instructed by the developer.

Do not delete database files as a first troubleshooting step.

Before Clearing Application Data

Always ask:

Has the latest successful backup been confirmed?

If the answer is no, do not clear the application data without first protecting the business data.

Device Security

Use Android's available security mechanisms such as:

- Screen lock
- Device PIN
- Secure physical storage
- Controlled staff access

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22. Troubleshooting

Problem: Application Does Not Open

Try:

1. Restart the device.
2. Open the application again.
3. Check available storage.
4. Check whether Android reports an application error.

If the problem continues, administrator/developer investigation may be required.

Problem: Products Are Missing

Check:

1. Database status.
2. Product list.
3. Correct application installation.
4. Whether products were saved successfully.

Do not immediately delete the database.

Problem: Barcode Does Not Scan

Check:

- Camera permission
- Camera lens
- Lighting
- Distance
- Barcode position
- Product barcode registration

Problem: Camera Is White or Unavailable

Check:

1. Camera permission.
2. Android camera availability.
3. Whether another application is using the camera.
4. Restart the SmartPOS if necessary.

Problem: Receipt Will Not Print

Check:

- Printer power
- Paper
- Network
- Printer selection
- Printer availability
- Printer configuration

Problem: M-Pesa STK Push Fails

Check:

- Internet connection
- Customer phone number
- M-Pesa configuration
- Transaction amount
- Returned payment result

Do not repeatedly initiate requests without checking whether an earlier request has already succeeded.

Problem: Backup Fails

Check:

1. Internet connection.
2. Backup configuration.
3. Available storage.
4. Database status.
5. Backup status/history.

Retry the backup.

Problem: Database Error

Do not immediately uninstall the application or clear data.

First:

1. Stop unnecessary activity.
2. Check whether a successful backup exists.
3. Restart the application.
4. Check database status.
5. Contact the administrator/developer if the problem continues.

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23. Maintenance

Regular maintenance helps keep the POS reliable.

Daily

- Check device power.
- Check printer.
- Check scanner.
- Review sales.
- Check backup status.

Weekly

- Review stock.
- Review sales reports.
- Review cashier activity.
- Check backup history.

Periodically

- Review device storage.
- Review Android updates.
- Check printer condition.
- Check camera condition.
- Confirm backup/recovery procedures.

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24. Recommended Business Procedures

Rule 1 — Record Every Sale

Every sale should go through the POS.

Rule 2 — Verify M-Pesa

Do not rely only on a customer's verbal confirmation.

Rule 3 — Protect Administrator Access

Only authorized personnel should have administrative access.

Rule 4 — Check Backups

A backup should be considered successful only after the system confirms it.

Rule 5 — Do Not Delete Data During Troubleshooting

Database deletion should never be the first solution.

Rule 6 — Reconcile Regularly

Compare:

POS records

against

Cash

and

M-Pesa/payment records.

Rule 7 — Protect the Device

The SmartPOS contains important business information and should be physically secured.

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25. Quick Reference Guide

Task	Procedure
Add Product	Product Management → Add Product → Enter details → Save
Scan Product	Scanner → Camera → Barcode → Product
Make Sale	Add Products → Review → Payment → Complete
M-Pesa	Phone → STK Push → Customer Confirms → Verify
Print Receipt	Completed Sale → Print → Select Printer
PDF Reference	Completed Sale → PDF/Reference
Check Finance	Finance/Reports → Review
Backup	Backup & Reports → Backup/Sync → Verify
Check Backup	Backup History → Last Successful Date
Stock	Product/Stock Management → Update Quantity

Task	Procedure
Cashier	Login → Sell → Receipt
Administrator	Login → Products/Stock/Reports/Backup

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26. Final Notes

The **Joskev Developers SmartPOS** is designed to provide a reliable and practical retail management environment.

Its most important operational principles are:

Local-first operation

The local SQLite database remains the primary working database.

Reliable sales

Sales should be recorded consistently and payment results should be verified.

Fast scanning

A good Android camera allows products to be identified quickly using barcodes.

M-Pesa integration

STK Push provides a convenient payment workflow while requiring proper transaction confirmation.

Professional receipts

Physical receipts and PDF sales references provide useful records for customers and management.

Business visibility

Finance and reports help administrators understand sales and revenue.

Backup protection

Automatic cloud backup provides an additional layer of protection while preserving the offline-first nature of the POS.

Recovery

If the device is offline for several days, local business activity remains available and pending backup information can be synchronized when connectivity returns.

Security

Administrator credentials, cashier accounts, the physical SmartPOS and local database should all be protected.

SmartPOS Operating Principle

SELL → RECORD → VERIFY → PRINT → BACK UP

The goal of the system is simple:

Keep the business selling, keep the records accurate, and keep the data protected.

Powered by Joskev Developers

SmartPOS — Retail Management • Sales • Stock • Payments • Reports • Backup